

8.2 Functions

HP-Z0000YPR-111005-02-862

n.a.

8.2.1 Dicom Services

HP-Z0000YPR-111005-02-854

Tools are provided for the DICOM Services **Verification, Storage, Storage Commitment, Query/Retrieve, Basic Print, Modality Worklist and Modality Performed Procedure Steps**. These tools may be selected under the menu item **DICOM Service**.

After you have selected a certain tool, you need to **Abort** it before invoking another tool or exiting the DICOM Test Tools.

Verify Network

Fig. 168: DICOM Verification of a single destination

DICOM Verify Network | Event Log | Help | Home | Analysis

DICOM Service: **Verify Network**

Verify all AETs: ☐

Timeout in sec:

Host Name: **SELECT**

Logical Name: **SELECT**

AET: **SELECT**

Info

Explanation: Checks the network connection to remote hosts and the DICOM accessibility of remote DICOM Application Entity Titles.

Action: Select a host from the **Host Name** selection list, then select a logical name from the **Logical Name** selection list and then select a Application Entity Title from the **AET** selection list. Alternatively you can select **Verify all AETs** to check all configured hosts and Application Entity Titles. You can tune the **Timeout** for establishing the connection between **1** and **1000** seconds. Then start the verification of the host(s) (ping) and Application Entity Title(s) (C-ECHO) with the **Go** button. The results will be shown in this frame.

Note: This may take some minutes.

Current time on remote system: 12/16/2000 09:47

DICOM Test Tools Verify Network

DICOM TestTools Verify Network ↔ TCP/IP Ping ↔ local or remote Host

DICOM TestTools Verify Network ↔ DICOM C-ECHO ↔ DICOM Service

| Go | Abort | Reset

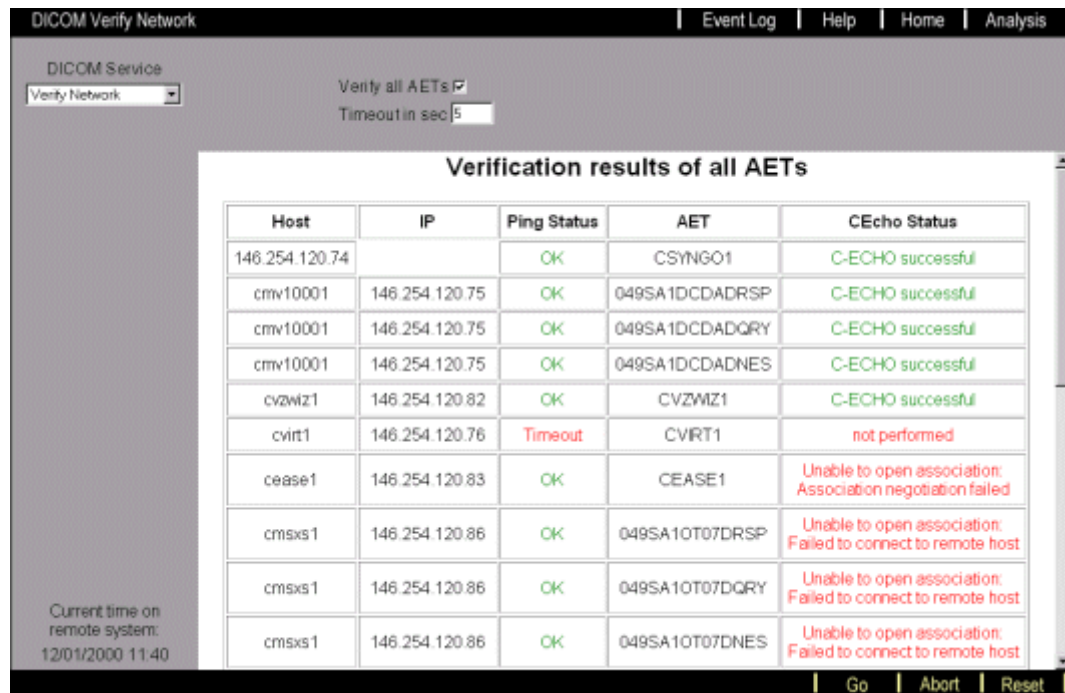
Verify Network performs a **TCP/IP ping**. If this ping is successful, the DICOM Service **DICOM Verification** sends a C-ECHO request to a DICOM destination process (AET) on the local or remote system.

DICOM Verification may be performed for a specific DICOM node or simultaneously for all configured DICOM nodes (Verify all AETs).

The test consists of 2 steps: The first step is the TCP/IP ping, and the second step is the C-ECHO command on the DICOM level.

You can stop the test by clicking the **Abort** button in the action bar.

Fig. 169: DICOM Verification of all AETs



- **Verify all AETs:**

A TCP/IP ping command and a DICOM C_ECHO command are performed automatically for all configured DICOM nodes. If a node is not running or not accessible, you have to wait for 30 seconds (default time).

- **Timeout in sec**

The default timeout for C_ECHO of 30 seconds may be changed here.

- **Host Name**

This field is only displayed if **Verify all AETs** is not selected. Select the host name of the destination device. The device itself (own host name) may be selected as well. After the logical name or AET (next two buttons) is selected, the appropriate host name will be displayed automatically.

- **Logical Name**

This field is only displayed if **Verify all AETs** is not selected. Select the logical name of the destination device. If the AET (next button) has been previously selected, the appropriate logical name will be displayed automatically.

- **AET**

Select the Application Entity Title of the destination device.

Troubleshooting remarks

- A TCP/IP ping may fail for the following reasons:
 - The remote host is turned off or not in the same network, no router is configured, or the routing demon is not started.
 - The TCP/IP address is configured incorrectly.
 - A networking device (router, switch, bridge, etc.) denies access to the other network.

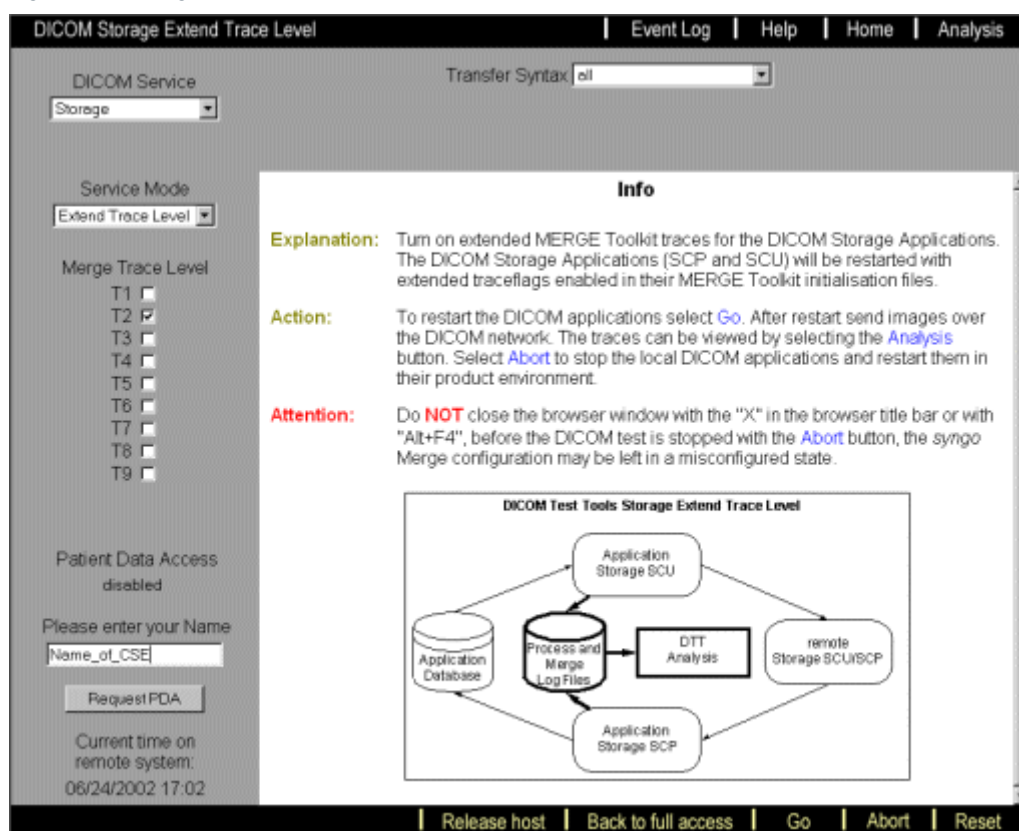
- A C-ECHO will fail for the following reasons:
 - AET or port number is not configured correctly.
 - The DICOM process on the remote host is not running.
 - Certain products check the IP address or AET of the sending system. If the AET or IP address of the local system is not entered correctly, DICOM verification will fail.
 - The remote host does not support DICOM verification as a Service Class Provider (see DICOM Conformance Statement).

Storage

The storage service class determines how DICOM images and other objects (waveforms, LUTs, overlays) are sent (SCU) or received (SCP). The DICOM Standard defines several transfer syntaxes, each of which is a set of encoding rules for communication (e.g. byte ordering, compression).

Extend Trace Level

Fig. 170: Storage - Extended trace menu



This DICOM implementation is based on a toolkit from Merge Technologies. This tool kit includes a mergecom.log file designed as a ring buffer. Certain trace levels may be enabled to receive more detailed information regarding the latest DICOM communications. The respective mergecom.log file can be displayed under Analysis at "Examine Logs - merge_stu.log". Basic knowledge of DICOM is necessary for interpreting the log files.

Storage, Storage Commitment and Query/Retrieve use the same mergecom.log file, as they belong to the same syngo service.

C:\%MEDHOME%\log\merge_stu.log Storage, Storage Commitment, and Query/Retrieve

The storage test tools may only be started remotely in “Full Access Mode”. In this case, no user interaction is possible. It is therefore not possible to send / print images or to perform a query request. Since these user interactions are required in the mergecom.log files to receive the appropriate messages, DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). It is even possible to iconize the “Remote Access Control Window”.

- **Back to full access**

After the images have been transferred or a Query/Retrieve has been performed, you have to disable interactions at the user interface to perform further tasks in the DICOM Test Tools.

- **Transfer syntax**

This drop-down list lets you choose transfer syntaxes used for the association negotiation after a restart of the DICOM processes. It provides additional troubleshooting possibilities, since you can limit the settings for the next associations. Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, “abort” the Extend Trace Level function. The original Transfer Syntaxes will be set again.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”. **Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.**

- **T1** This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.



Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems on the T3 level are also displayed.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

When incoming associations are automatically rejected (IP address not known), log messages will be displayed at this trace level.

- **T5 regular and extended regulations**

Logs messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, although it requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image.

The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed on the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line. The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) is activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Send Images

This menu item can be used to send DICOM images (stored on the hard disk via the Test Receiver or, e.g. FTP) to a user-specified DICOM receiver. By sending these images to the local DICOM receiver, you are able to import images into the database and display them in the user interface.

Fig. 171: Storage - Send Images menu

DICOM Storage Send Images | Event Log | Help | Home | Analysis

DICOM Service: **Storage** | Transfer Syntax: **all** | Host Name: **cvirt1**
 Image Data: **SELECT** | Logical Name: **CCC_Virtuoso**
 AET of Sender: **CSYNG01** | AET: **CVIRT1**

Service Mode: **Send Images**

Merge Trace Level:
 T1 ☐ T2 ☒ T3 ☒ T4 ☐
 T5 ☐ T6 ☐ T7 ☒ T8 ☐ T9 ☐

Current time on remote system:
 12/16/2000 09:57

Info

Explanation: Send images to remote DICOM Storage SCP's. The images you can send are stored in the images directory of the DICOM Service Tools. They are stored there by the DICOM Storage Test Receiver or by a user action.

Action: Please select the Application Entity Title of the remote DICOM Storage SCP from the **Application Entity** selection list and the file to be send from the **Image** selection list. Start the image transfer with the **Go** button. The **Analysis** button activates the **Analyse Tools** menu.

DICOM Test Tools Storage Send Image

```

graph LR
    AD[(Application Database)] --> ASCU[Application Storage SCU]
    HDD[(Harddisk DTT Image Directory)] --> DTTSCU[DICOM Test Tools Storage SCU]
    ASCU --> RSCP[remote Storage SCP]
    DTTSCU --> RSCP
  
```

| Go | Abort | Reset |

■ Transfer syntax

This drop-down list lets you choose transfer syntaxes that will be enabled after the DICOM processes are restarted. It provides additional troubleshooting possibilities, as you can limit the settings for the next associations.

Before selecting other DICOM Services or functions, or before finishing the DICOM Test Tools, "abort" the "Send Images" function. The original Transfer Syntaxes will be set again.

■ Image Data

A list of DICOM images located in the directory %MEDHOME%\service\dstlimg is provided. The Test Receivers (see Simulate/Add local/remote Receiver) will copy all received images into this directory. Images copied manually (ftp) into this directory will also be displayed. Certain default DICOM images will be delivered with the DICOM Test Tools for test purposes.

■ AET of Sender

The sending AET is, by default, the AET of the local storage SCP. Usually it is not necessary to change this value.

■ Host Name

Host name of the destination system. The system itself (host name) may be selected as well. When selecting the logical name or AET (next two buttons), the appropriate host name will be displayed automatically.

- **Logical Name**

Select the logical name of the destination device. When selecting the AET (next button) first, the appropriate logical name will be displayed automatically.

- **AET**

Select the Storage AET of the destination device. Please note: The Send function will fail if you e.g. select the Query/Retrieve AET of the remote system instead of the Storage AET.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under "Analysis - Examine Logs".

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.



Trace level T2 also shows patient data in the log file. For this reason, "Patient Data Access" is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Logs messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, please note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image.

The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

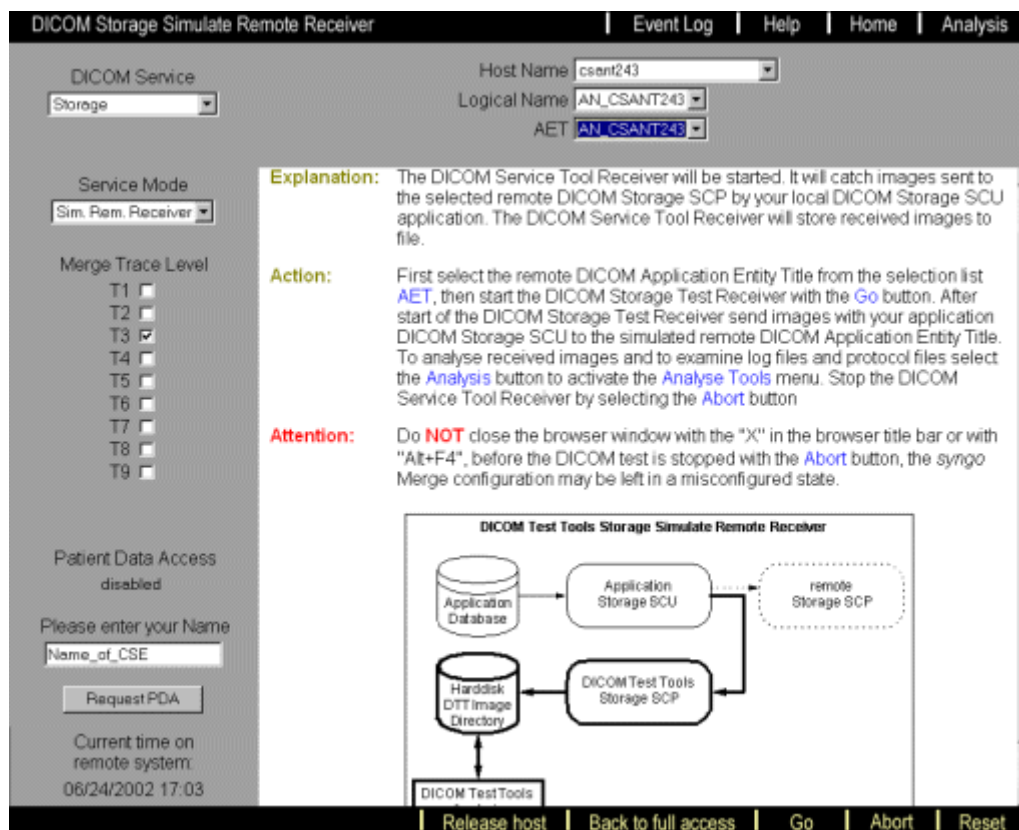
- **Request PDA**

Selecting this button initiates the activation of an additional selection item (Patient Data Access) in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Before selecting other DICOM Services or functions, or before finishing the DICOM Test Tools, "abort" the "Send Images" function.

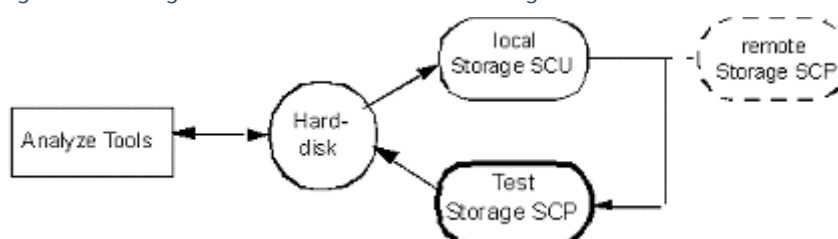
Simulate Remote Receiver

Fig. 172: Storage - Simulate Remote Receiver menu



When this menu item is selected, the Test Receiver provided by the DICOM Test Tools is started to simulate a remote DICOM receiver. You will receive a list of all DICOM AETs configured on your system. Select the AET you would like to have replaced with the Test Receiver. All images sent by the local sender to this destination will be received by the Test Receiver and stored on the hard disk in DICOM format.

Fig. 173: Storage - Simulate Remote Receiver diagram



Go to Analysis to have the received images evaluated. An image header dump may be performed, or the image can be displayed.

The received images and the log files may be removed when exiting the DICOM Service Tools.

The storage test tools may only be started remotely in "Full Access Mode". In this case, no user interaction is possible, and, therefore, it is also not possible to send / print images or to perform a query request. Since these user interactions are necessary to receive the appropriate messages in the mergecom.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). It is even possible to iconize the "Remote Access Control Window".

- **Back to full access**

After the images have been transferred or a Query/Retrieve has been performed, this function disables interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Host Name**

Host name of the destination system. The system itself (host name) may be selected as well. When selecting the logical name or AET (next two buttons), the appropriate host name will be displayed automatically.

- **Logical Name**

Select the logical name of the destination device. When selecting the AET (next button) first, the appropriate logical name will be displayed automatically.

- **AET**

Select the Storage AET of the destination device. Please note: The Send function will fail if you e.g. select the Query/Retrieve AET of the remote system instead of the Storage AET.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under "Analysis - Examine Logs".

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.



Trace level T2 also shows patient data in the log file. For this reason, "Patient Data Access" is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, please note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

The objects (images) captured with the Test Receiver will normally be anonymized. If patient-relevant data is required for troubleshooting (not anonymized), it is necessary to obtain the "Patient Data Access" rights from the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

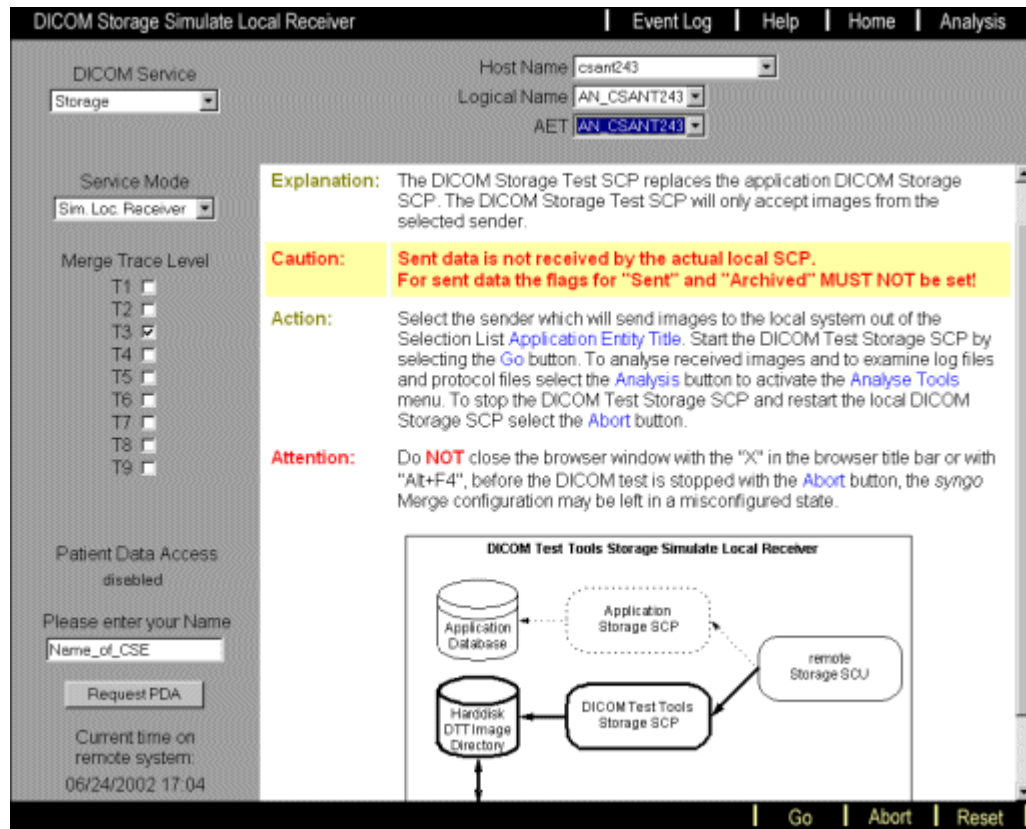
- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, "abort" the "Simulate Remote Receiver" function.

Simulate Local Receiver

Fig. 174: Storage - Simulate Local Receiver menu

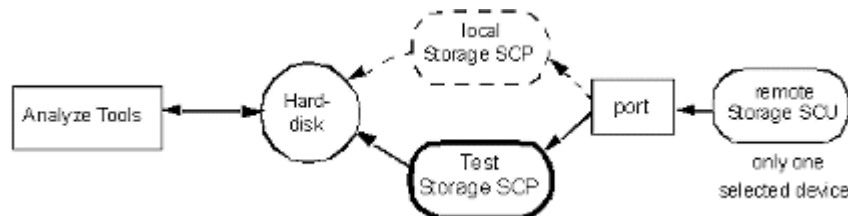


When this menu item is selected, a Test Receiver provided by the DICOM Test Tools is started using the same AET and port number as the application DICOM receiver. The original application receiver will be stopped. All images sent by remote senders will be stored in DICOM format on the hard disk.



None of the images received by the Test Receiver will be stored in the database. To avoid losing images, select the AET of the sending system. All other senders will be rejected.

Fig. 175: Storage - Simulate Local Receiver diagram



The received images may be analyzed with the tools provided by "DICOM Analysis".

The received images and the log files may be removed when exiting from the DICOM Service Tools.

■ Host Name

Host name of the sending system. For data security, only images from this AET will be received by the Test Receiver. All other systems will be rejected. The system itself

(host name) may be selected as well. When selecting the logical name or AET (next two buttons), the appropriate host name will be displayed automatically.

- **Logical Name**

Select the logical name of the sending system. When selecting the AET (next button) first, the appropriate logical name will be displayed automatically.

- **AET**

Select the storage AET of the sending system. Please note: The Send function will fail if you e.g. select the Query/Retrieve AET of the remote system instead of the Storage AET.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under "Analysis - Examine Logs".

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.



Trace level T2 also shows patient data in the log file. For this reason, "Patient Data Access" is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, please note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

The objects (images) captured with the Test Receiver will normally be anonymized. If patient-relevant data is required for troubleshooting (not anonymized), it is necessary to obtain the "Patient Data Access" rights from the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

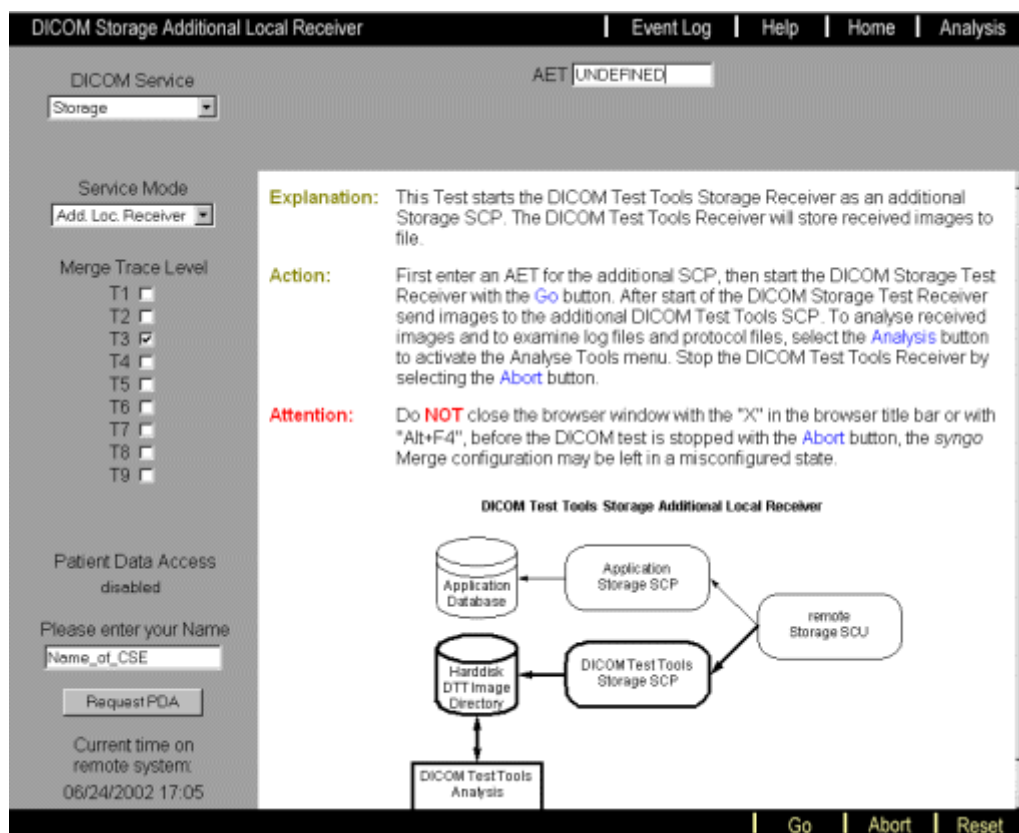
- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) is activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, "abort" the "Simulate Remote Receiver" function.

Additional Local Receiver

Fig. 176: Storage - Additional Local Receiver menu



This option allows you to select an additional DICOM receiver that is used for test purposes only. The AET of this Test Receiver has to be entered during configuration. The Test Receiver is listening under port **110**. The Test Receiver may be used to receive images from the local application as well as any remote system (as long as this AET is configured there).

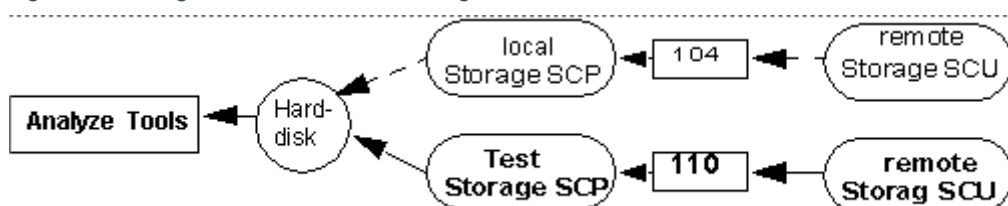
You can only send from a certain AET, which needs to be selected. All other remote nodes can still send to the original DICOM receiver.



None of the images received by the Test Receiver will be stored in the database. To avoid losing images, select the AET of the sending system. All other senders will still send to the original local DICOM receiver.

To send to the additional Test Receiver, configure this Test Storage SCP (port: 110, AET will be defined in the test tools) at the remote system (SCU).

Fig. 177: Storage - Add Local Receiver diagram



The received images may be analyzed with the tools provided by “DICOM Analysis”.

The images received and the log files may be removed when exiting the DICOM Service Tools.

- **AET**

You need to define an AET for the additional Test Receiver. At the remote system, you need to configure this AET with port number 110.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”.

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.



Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system SCP. The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, please note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending, for example, an image, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

The objects (images) captured with the Test Receiver will normally be anonymized. If patient-relevant data is required for troubleshooting (not anonymized), it is necessary to obtain the "Patient Data Access" rights from the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

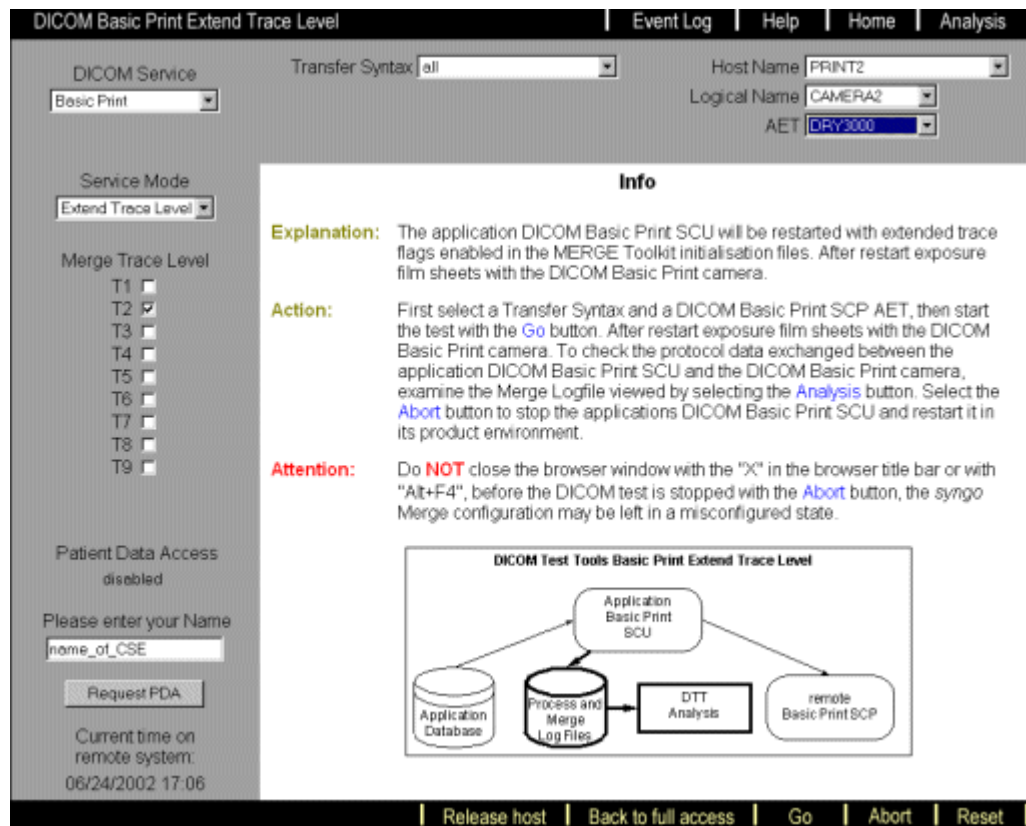
Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, it is necessary to "abort" the "Simulate Remote Receiver" function.

Basic Print

This menu provides tools for testing the Basic Print function of the local DICOM Basic Print process. You may start a Test Basic Print provider which replaces the remote Basic Print provider (usually a laser camera). After printing a film from the user interface to the Test Basic Print process, you can analyze the log files and display the film using the tools provided under Analysis.

Extend Trace Level

Fig. 178: Basic Print - Extended trace menu



This DICOM implementation is based on a toolkit from Merge Technologies. This tool kit includes a mergecom.log file designed as a ring buffer. Certain trace levels may be enabled to receive more detailed information regarding the latest DICOM communications. The respective mergecom.log file can be displayed under Analysis at "Examine Logs - merge_hc.log". Basic knowledge of DICOM is necessary for interpreting the log files.

Location and name of the mergecom.log file for Basic Print:

C:\%MEDHOME%\log\merge_hc.log Basic Print (hardcopy)

The Print Test Tools may only be started remotely in "Full Access Mode". In this case, no user interaction is possible, and, therefore, it is also not possible to send / print images or to perform a query request. Since these user interactions are necessary to receive the appropriate messages in the mergecom.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). The "Remote Access Control Window" can be iconized.

- **Back to full access**

After the images have been sent to the camera, disable interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Transfer syntax**

This drop-down list allows you to choose transfer syntaxes that will be used for the association negotiation after a restart of the DICOM processes. It provides additional

troubleshooting possibilities, as you can limit the settings for the next associations. Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, "abort" the Extend Trace Level function. The original Transfer Syntaxes will be set again.

- **Host Name**

Host name of the camera. The system itself (host name) may be selected as well. When selecting the logical name or AET (next two buttons), the appropriate host name will be displayed automatically.

- **Logical Name**

Select the logical name of the camera. When selecting the AET (next button) first, the appropriate logical name will be displayed automatically.

- **AET**

Select the AET of the camera.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under "Analysis - Examine Logs". **Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.**

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.



Trace level T2 also shows patient data in the log file. For this reason, "Patient Data Access" is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known); log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

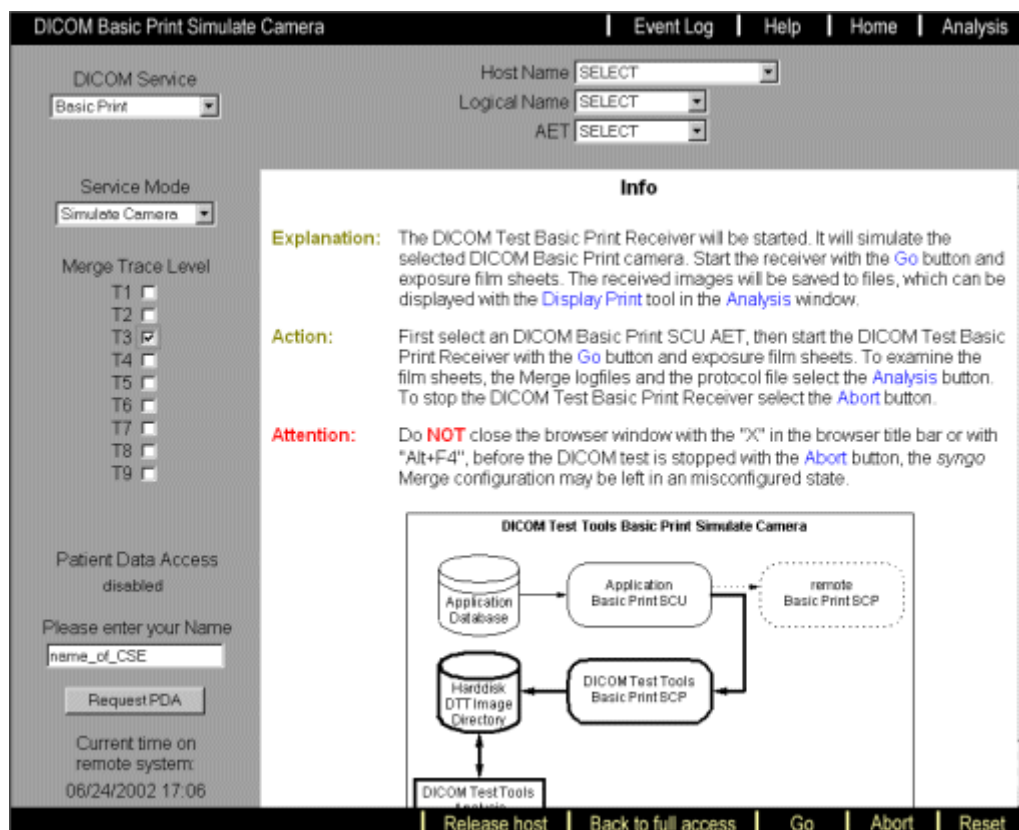
Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

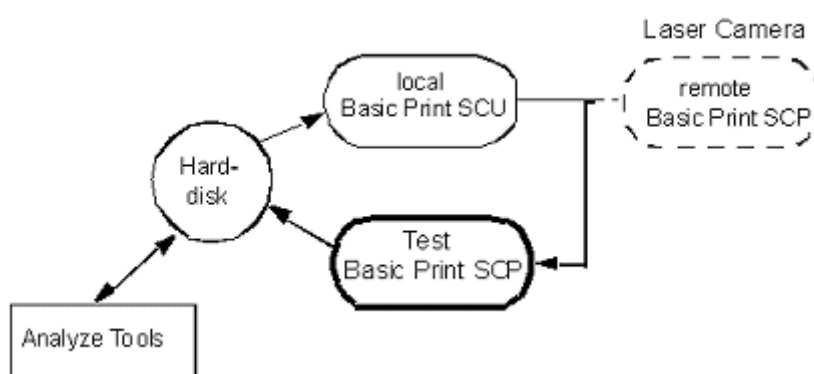
Simulate Camera

Fig. 179: Basic Print - Simulate Camera menu



When this menu item is selected, the Test Basic Print SCP provided by the DICOM Service Tools is started in place of the camera (remote Basic Print SCP). You will receive a list of all configured Basic Print AETs. Select the appropriate laser camera. All films sent from the user interface to the corresponding laser camera will then be received by the Test Basic Print SCP.

Fig. 180: Basic Print - Simulate Camera diagram



The Test Basic Print SCP is not identical to the laser camera. As a result, its behavior may be different from the remote Basic Print SCP.

The Print Test Tools may only be started remotely in “Full Access Mode”. In this case, no user interaction is possible, and, therefore, it is also not possible to send / print images or to perform a query request. Since these user interactions are necessary to receive the appropriate messages in the mergecom.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). It is even possible to iconize the “Remote Access Control Window”.

- **Back to full access**

After the images have been sent to the camera, disable interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Host Name**

Host name of the camera. The system itself (host name) may be selected as well. When selecting the logical name or AET (next two buttons), the appropriate host name will be displayed automatically.

- **Logical Name**

Select the logical name of the camera. When selecting the AET (next button) first, the appropriate logical name will be displayed automatically.

- **AET**

Select the AET of the camera.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”. **Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.**

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.



Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access." Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

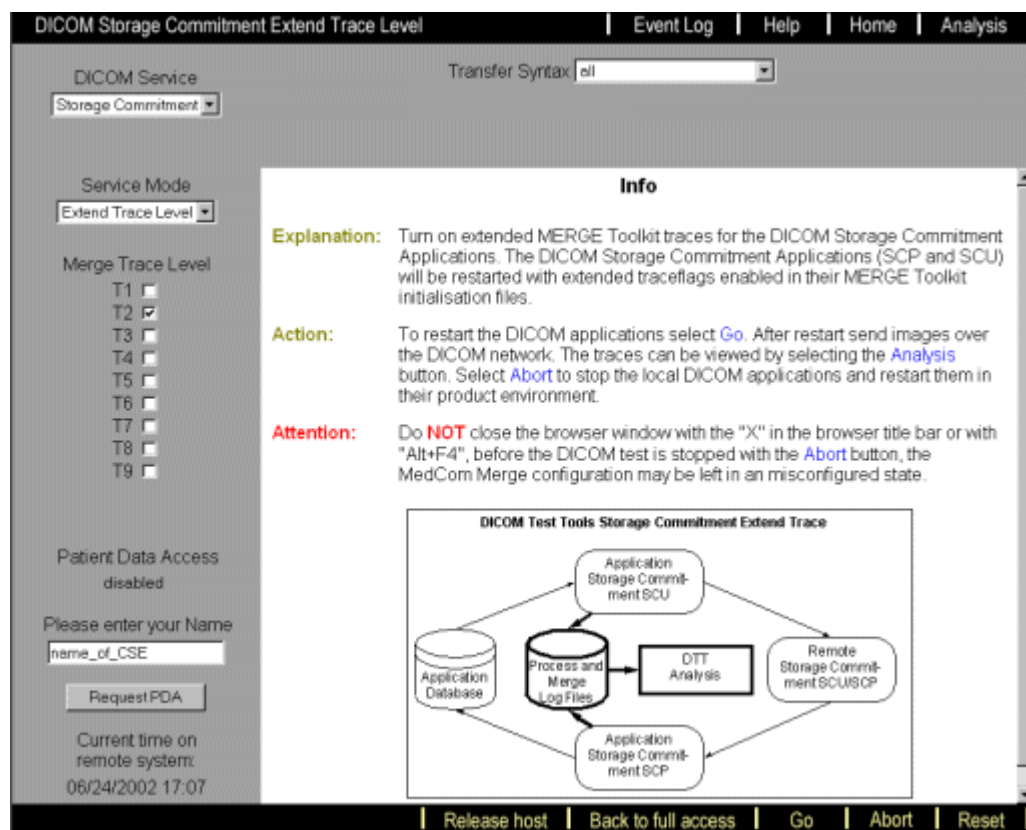
Storage commitment

Storage Commitment sends a list of image UIDs to the configured destination and waits for the response of the remote provider.

For this service, it is only possible to extend the trace level of the appropriate mergecom.log file. Storage Commitment uses the same DICOM process as Storage and therefore also the same log file, which is stu_merge.log. In this response, trace level T2 shows whether or not the images have been successfully stored at the destination.

Extend Trace Level

Fig. 181: Storage Commitment - Extend trace menu



This DICOM implementation is based on a toolkit from Merge Technologies. This tool kit includes a mergecom.log file designed as a ring buffer. Certain trace levels may be enabled to obtain more detailed information regarding the latest DICOM communications. The respective mergecom.log file can be displayed under Analysis at "Examine Logs - merge_stu.log". Basic knowledge of DICOM is necessary for interpreting the log files.

Storage, Storage Commitment and Query/Retrieve use the same mergecom.log file, as they belong to the same syngo service.

C:\%MEDHOME%\log\merge_stu. logs Storage, Storage Commitment and Query/Retrieve

The DICOM Test Tools may only be started remotely in "Full Access Mode". In this case, no user interaction is possible, and, therefore, it is also not possible to send / print images or to perform a query request. Since these user interactions are necessary to receive the appropriate messages in the mergecom.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

■ Release host

Select this item in the Action bar to work with the user interface (identical to limited access). It is even possible to iconize the "Remote Access Control Window".

- **Back to full access**

After the images have been transferred or a Query/Retrieve has been performed, this function disables interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Transfer syntax**

This drop-down list allows you to choose transfer syntaxes that will be used for the association negotiation after a restart of the DICOM processes. It provides additional troubleshooting possibilities, as you can limit the settings for the next associations.

Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, "abort" the Extend Trace Level function. The original Transfer Syntaxes will be set again.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under "Analysis - Examine Logs".

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.



Trace level T2 also shows patient data in the log file. For this reason, "Patient Data Access" is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each

image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

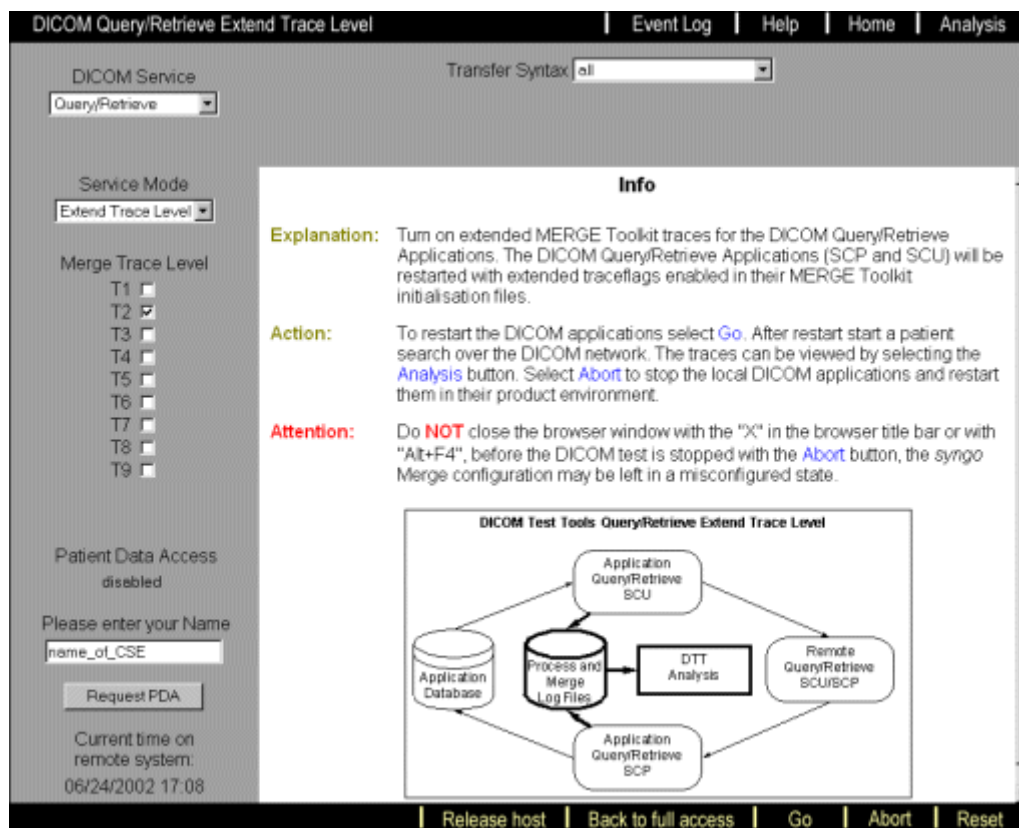
When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Query/Retrieve

For this service, it is only possible to extend the trace level of the appropriate merge-com.log file. Query/Retrieve uses the same DICOM process as Storage and therefore also the same log file, which is stu_merge.log. The Query request and response message attributes may be displayed on trace level T2.

Extend Trace Level

Fig. 182: Query/Retrieve - Extend trace menu



This DICOM implementation is based on a toolkit from Merge Technologies. This tool kit includes a mergecom.log file designed as a ring buffer. Certain trace levels may be enabled to obtain more detailed information regarding the latest DICOM communications. The respective mergecom.log file can be displayed under Analysis at "Examine Logs - merge_stu.log". Basic knowledge of DICOM is necessary for interpreting the log files.

Storage, Storage Commitment and Query/Retrieve use the same mergecom.log file, as they belong to the same syngo service.

C:\%MEDHOME%\log\merge_stu. logs Storage, Storage Commitment and Query/Retrieve

The DICOM Test Tools may only be started remotely in "Full Access Mode". In this case, no user interaction is possible, and, therefore, it is also not possible to send / print images or to perform a query request. Since these user interactions are necessary to receive the appropriate messages in the mergecom.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). The "Remote Access Control Window" can be iconized.

- **Back to full access**

After the images have been transferred or a Query/Retrieve has been performed, this function disables interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Transfer syntax**

This drop-down list allows you to choose transfer syntaxes that will be used for the association negotiation after a restart of the DICOM processes. It provides additional troubleshooting possibilities, as you can limit the settings for the next associations.

Before selecting other DICOM services or functions, or before finishing the DICOM Test Tools, “abort” the Extend Trace Level function. The original Transfer Syntaxes will be set again.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”.

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.



Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

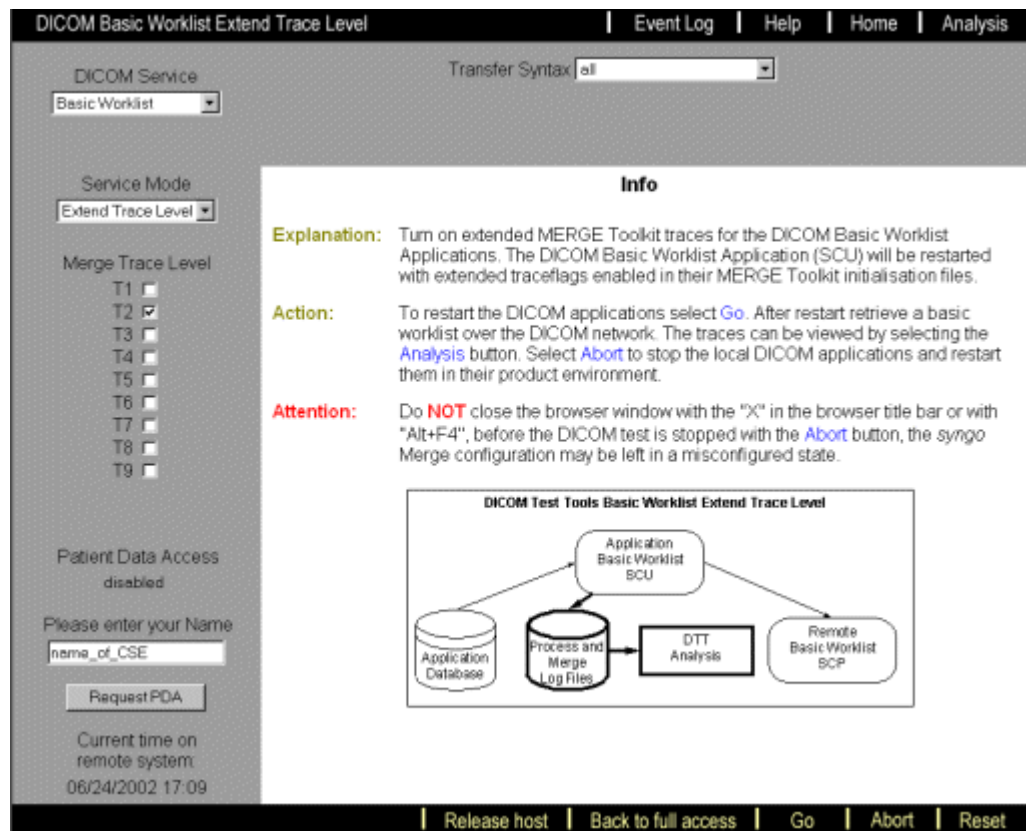
When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Basic Worklist

For this service, it is only possible to extend the trace level of the appropriate merge-com.log file. The Worklist request and response message attributes may be displayed at trace level T2.

Extend Trace Level

Fig. 183: Basic Worklist - Extend trace menu



This DICOM implementation is based on a toolkit from Merge Technologies. This tool kit includes a mergecom.log file designed as a ring buffer. Certain trace levels may be enabled to obtain more detailed information regarding the latest DICOM communications. The respective mergecom.log file can be displayed under Analysis at "Examine Logs - merge_wl.log". Basic knowledge of DICOM is necessary for interpreting the log files.

The DICOM Test Tools may only be started remotely in "Full Access Mode". In this case, no user interaction is possible. It is therefore not possible to perform a Worklist request from the user interface. Since user interaction is necessary to receive the appropriate messages in the merge_HRI.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). It is even possible to iconize the "Remote Access Control Window".

- **Back to full access**

After the Worklist request has been performed, disable interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Transfer syntax**

This drop-down list allows you to choose transfer syntaxes that will be used for the association negotiation after a restart of the DICOM processes. It provides additional troubleshooting possibilities, as you can limit the settings for the next associations. Before selecting other DICOM services or functions, or before finishing the DICOM

Test Tools, “abort” the Extend Trace Level function. The original Transfer Syntaxes will be set again.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”.

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.



Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means a successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

If the DICOM Service Tools are accessed remotely, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

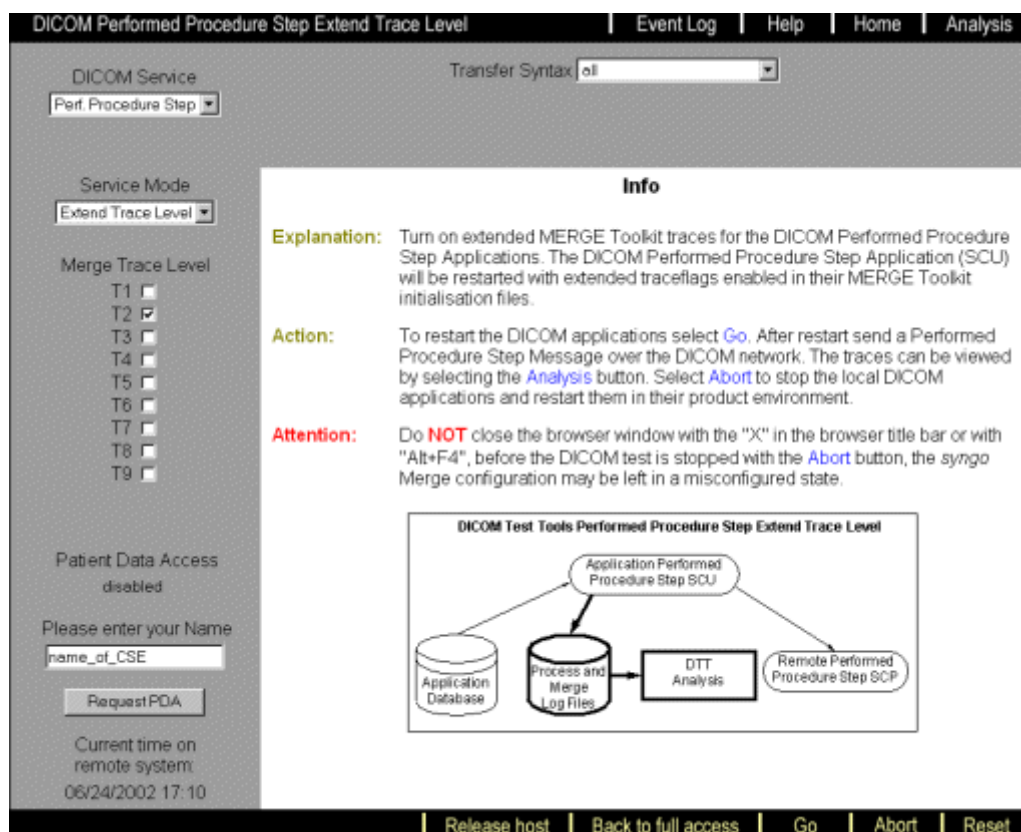
When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

Modality Performed Procedure Step

For this service, it is only possible to extend the trace level of the appropriate merge-com.log file. The Worklist request and response message attributes may be displayed at trace level T2.

Extend Trace Level

Fig. 184: Modality Performed Procedure Step - Extend trace menu



This DICOM implementation is based on a toolkit from Merge Technologies. This tool kit includes a mergecom.log file designed as a ring buffer. Certain trace levels may be enabled to obtain more detailed information regarding the latest DICOM communications. The respective mergecom.log file can be displayed under Analysis at "Examine Logs - merge_HRI.log". Basic knowledge of DICOM is necessary for interpreting the log files.

The DICOM Test Tools may only be started remotely in "Full Access Mode". In this case, no user interaction is possible. It is therefore not possible to perform a Worklist request from the user interface. Since user interaction is necessary to receive the appropriate messages in the merge_HRI.log files, the DICOM Test Tools provide the option of enabling user interaction even in full access mode.

- **Release host**

Select this item in the Action bar to work with the user interface (identical to limited access). It is even possible to iconize the "Remote Access Control Window".

- **Back to full access**

After the Worklist request has been performed, disable interaction at the user interface to perform further tasks in the DICOM Test Tools.

- **Transfer syntax**

This drop-down list allows you to choose transfer syntaxes that will be used for the association negotiation after a restart of the DICOM processes. It provides additional troubleshooting possibilities, as you can limit the settings for the next associations. Before selecting other DICOM services or functions, or before finishing the DICOM

Test Tools, “abort” the Extend Trace Level function. The original Transfer Syntaxes will be set again.

- **Merge trace level**

The following trace levels may be enabled to provide more detailed information in the log file merge_stu.log. This log file may be displayed under “Analysis - Examine Logs”.

Also refer to the example of an association with trace level T2, T3, T7 and T9 in the appendix.

- **T1**

This trace level does not affect the mergecom.log file.

- **T2 data set**

This trace level shows the attributes of the transferred message, e.g. the image header attributes or the attributes within a DICOM message.



Trace level T2 also shows patient data in the log file. For this reason, “Patient Data Access” is required for remote access.

- **T3 association negotiation**

Every DICOM communication starts with the association negotiation. The initiator (SCU) sends a list of all supported image objects and transfer syntaxes to the remote system (SCP). The AETs of both systems are also displayed at the T3 level.

The SCP responds with the status (accept or reject) of the image objects and selects one of the proposed transfer syntaxes.

- **T4 rejection of incoming messages**

If incoming associations are automatically rejected (IP address not known), log messages will be displayed on this trace level.

- **T5 regular and extended regulations**

Log messages relating to regular and extended regulations.

- **T6 mergecom configuration files**

At this trace level the related mergecom.ini, mergecom.pro, mergecom.app, and mergecom.srv files will be written in the mergecom.log file. This may be helpful when analyzing the log file, however, note that analysis requires a large amount of space in the log file.

- **T7 command set**

Following the association negotiation, a command set (attributes with group 0) will be transferred. Before sending an image, for example, the command set will tell the remote system the type of image object (SOP Class UID such as CT, RF, etc.) to be transferred, including the UID of the image. The response command set after each image will show the status of the transfer, where status = 0 means successful transfer.

- **T8 streaming messages**

Log messages relating to the streaming in and out of messages and file objects are displayed at the T8 level.

- **T9 PDU level**

The entire DICOM association is displayed on the PDU level (Protocol Data Unit). Each PDU defined with a maximum size during the association negotiation will be displayed in one line.

The association negotiation will be shown in greater detail.

T9 is the only way to display the association release request and response at the end of an association.

Patient Data Access

In case of remote access to the DICOM Service Tools, viewing patient-relevant data (name, ID, etc.) is not permitted. However, if you enable trace level T2, patient data will be displayed in the log file. For this purpose, the customer has to provide additional access rights, allowing the service engineer to view patient-relevant data. To obtain these rights, the service engineer has to enter his name and send a "Patient Data Access" request to the customer's user interface.

- **Please enter your name**

Enter the name of the local service engineer logged in remotely who requires Patient Data Access rights.

- **Request PDA**

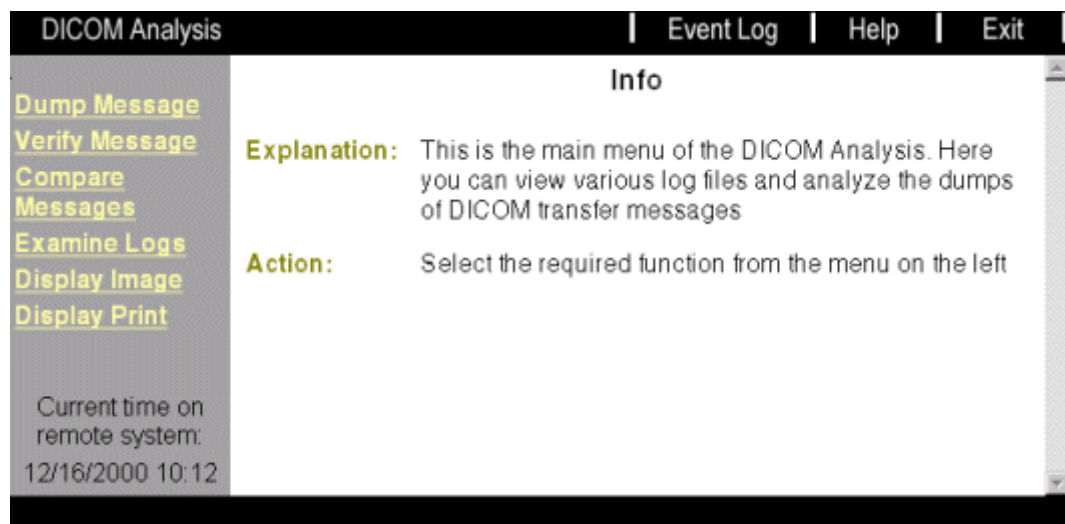
When you select this button, an additional selection item (Patient Data Access) will be activated in the customer user interface under "Option - Remote Access". Additionally, the name of the service engineer will be displayed. The customer has to select this menu item for trace level T2 to be activated in the log file.

8.2.2 Dicom Analysis

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The analysis tools provide the option of displaying and evaluating the messages and images captured by the Test Receiver (Storage, Basic Print) as well as displaying the appropriate mergecom.log files.

Fig. 185: Dicom Analysis menu



Dump Message

The image header information of images captured by the Test Receiver (simulate local/remote receiver), sent via "Export to Offline" from the customer user interface or copied manually into the directory %MEDHOME%\service\dst\img will be displayed by different header dump tools. The image has to be selected from a given selection list. Please note that all saved files may be removed when exiting the DICOM Test Tools (selectable).

Fig. 186: Dicom Dump Message mc3list

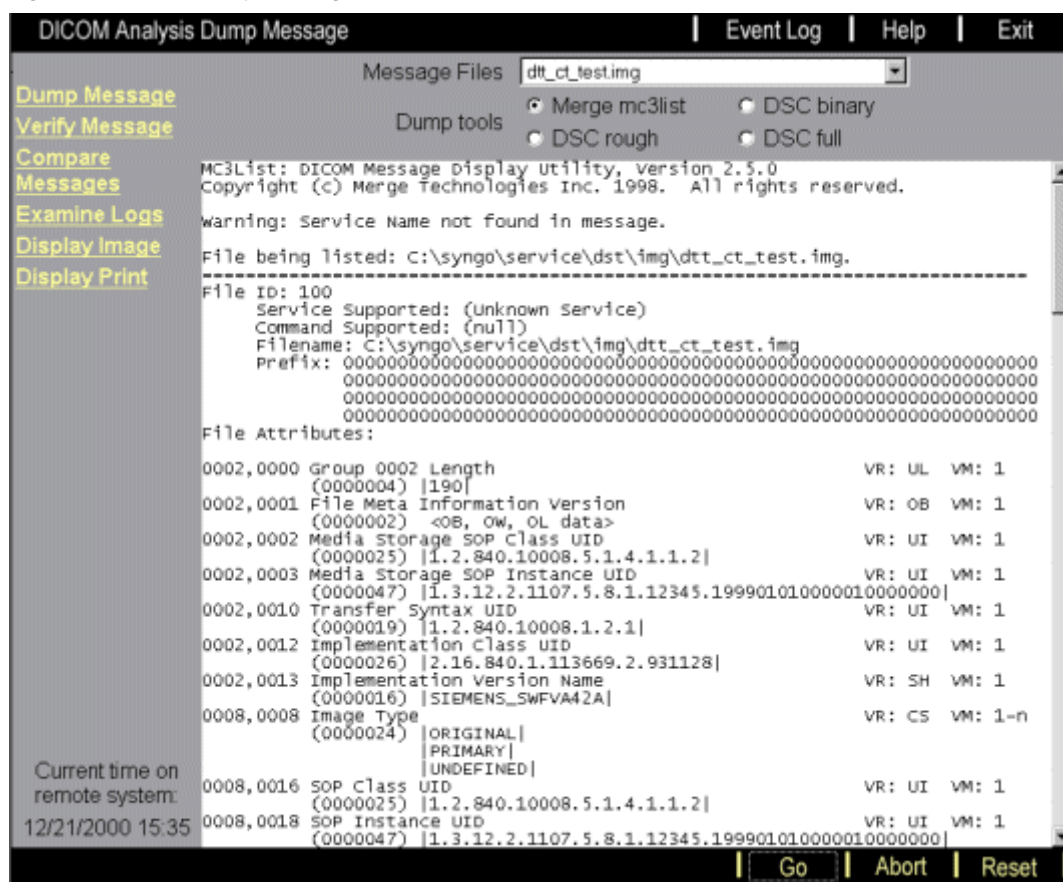


Fig. 187: Dicom Dump Message binary

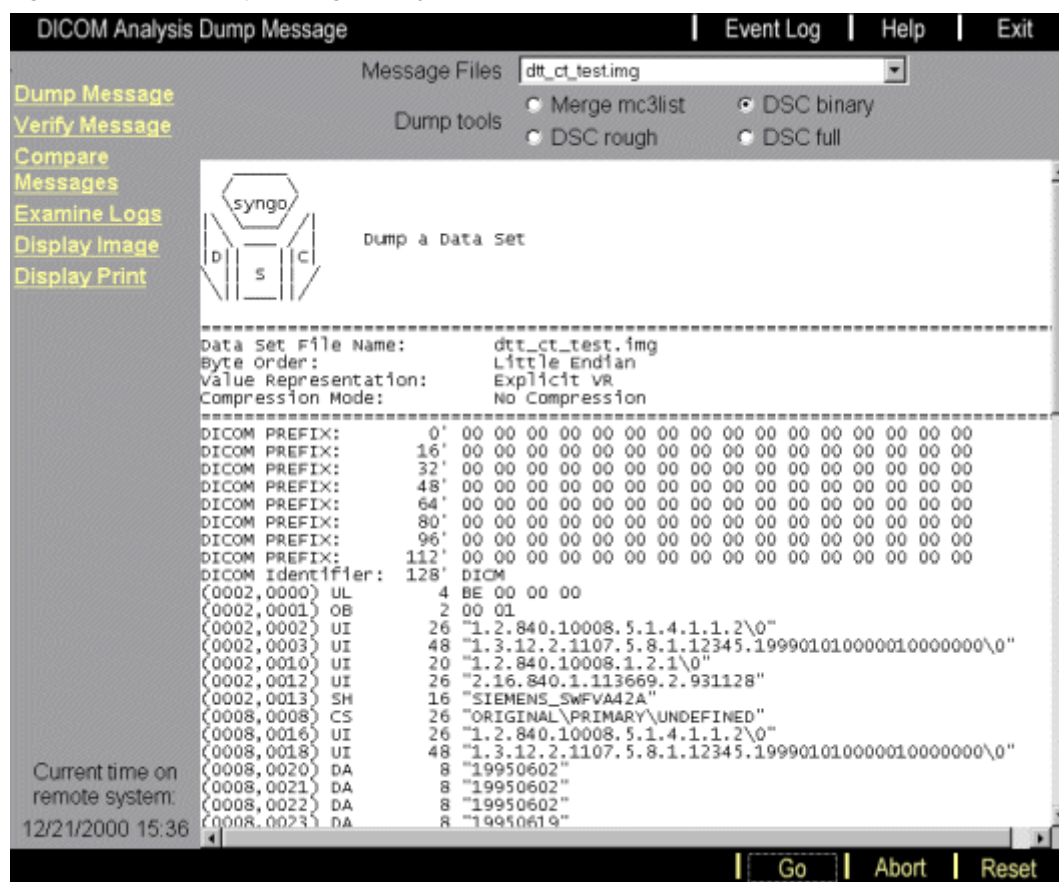
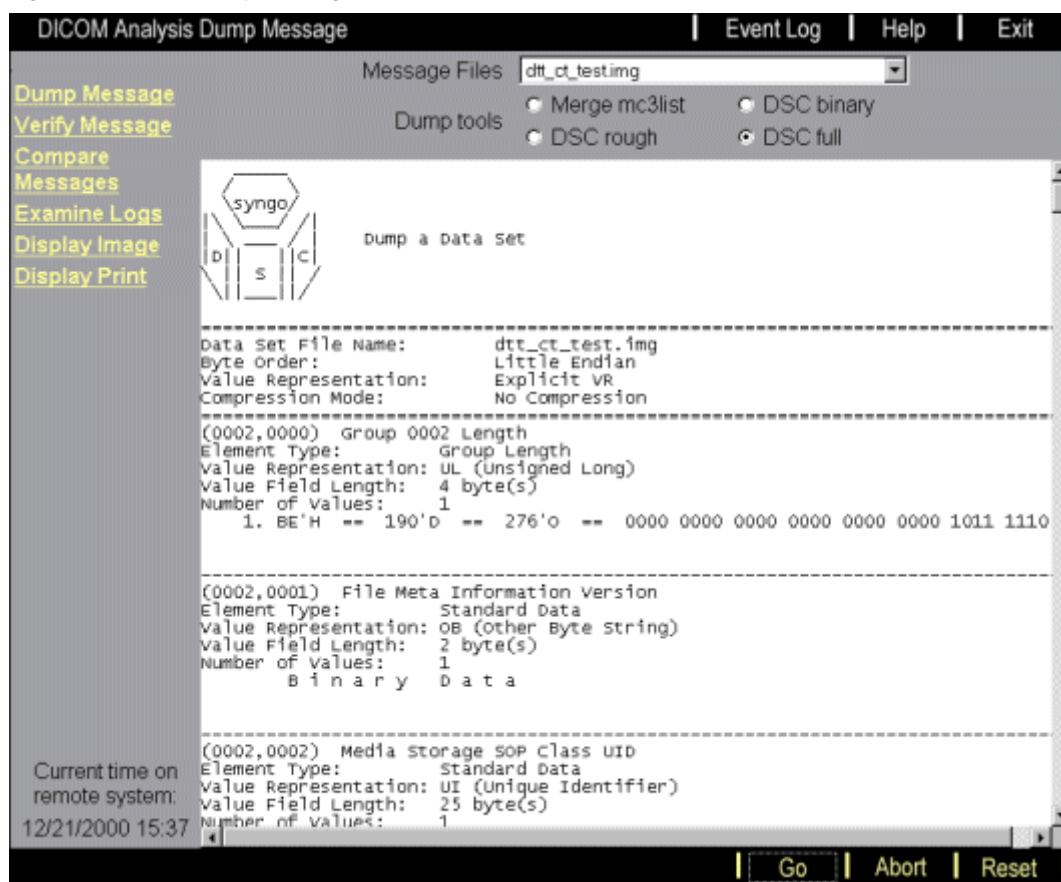


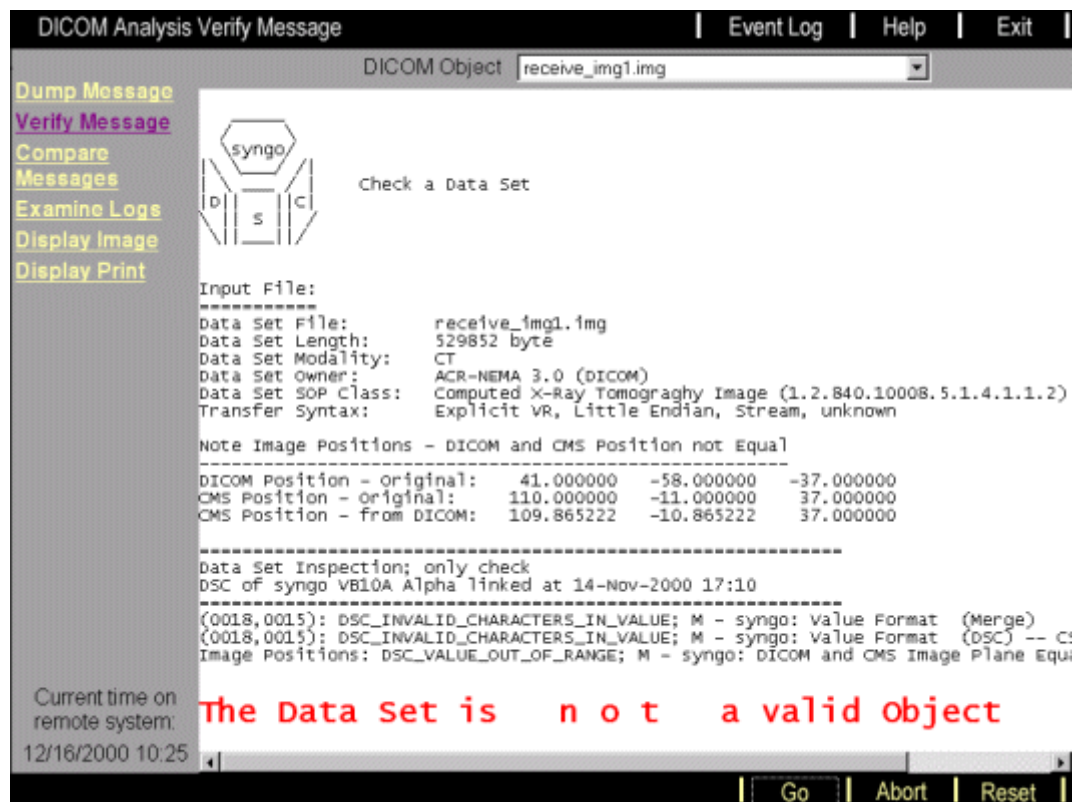
Fig. 188: Dicom Dump Message full



Verify Message

The image header information of images captured by the Test Receiver (simulate local/remote receiver) and sent via "Export to Offline" from the customer user interface or copied manually to the %MEDHOME%\service\dst\img directory will be evaluated for DICOM conformity. Please note that not all image header attributes marked "Error" are in violation of the DICOM Standard. For example, missing or incorrect attributes will be displayed even if they are optional attributes. The tool provides a means of identifying the nature of such "Error" messages. Each indicated header attribute should also be verified manually using the DICOM Standard.

Fig. 189: Dicom Verify Message menu



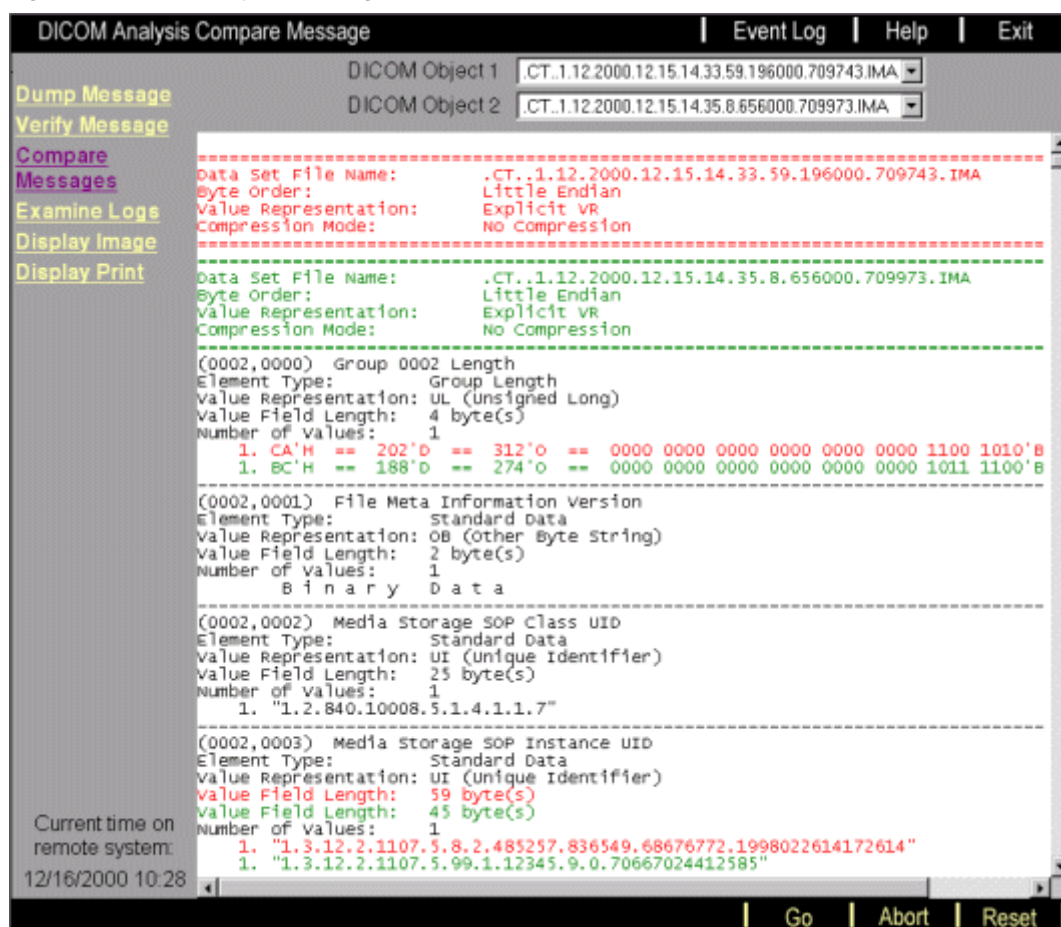
Compare Message

Two images captured by the Test Receiver (simulate local/remote receiver) and sent via "Export to Offline" from the customer user interface or copied manually to the %MEDHOME%\service\dst\img directory will be compared. The message has to be selected by the user from a given selection list.

This example shows a possible test scenario:

- Send an image to a remote DICOM node.
- Simulate the remote node and resend the same image. This image will be stored in DICOM format on the hard disk.
- Simulate the local receiver and send the image from the remote system to the local system. This image will also be stored in DICOM format on the hard disk.
- Compare the sent image with the received image.

Fig. 190: Dicom Compare Message menu



Examine Logs

This menu item displays the log directory including the mergecom.log files of the appropriate DICOM Services. Choose the process for which you want to see the log file:

Merge log files

Application Storage SCU/SCP	merge_stu.log
Test Tools Storage SCU	merge_dst_sscu.log (process log: dst_sscu_out.log)
Test Tools Storage SCP	merge_dst_sscp.log (process log: dst_sscp_out.log)
Application Storage Commitment SCU	merge_stu.log
Application Query/Retrieve SCU	merge_stu.log
Application Basic Print SCU	merge_hc.log
Test Tools Basic Print SCP	merge_dst_pscp.log (process log: dst_pscp_out.log)

Application Worklist SCU

[merge_HRI.log](#)Application Modality Performed Procedure
Step SCU[merge_HRI.log](#)

The log file is displayed when you click the appropriate file name.

Fig. 191: Dicom Examine Logs menu



Display Image

DICOM images captured by the Test Receiver (simulate local/remote receiver) and sent via "Export to Offline" from the customer user interface or copied manually to the %MEDHOME%\service\dst\img directory will be displayed. The image has to be selected by the user from a given selection list.



The viewer is a Java applet that **does not support all types of DICOM images**. If the images cannot be displayed properly, you need to transfer them to your PC and display them with other viewer applications such as the MV300 Viewer.

The following images cannot be displayed:

- Monochrome 16 bit images with bits of 8, 10, 15, 16 stored
- Palette Color images
- Multiframe images - only the frame in the middle of the sequence will be displayed
- images that do not have the attributes of group 2 (part 10 header for offline images)

The following images are displayed:

- Monochrome 8 bit images and 16 bit images with bits of 12 stored
- RGB images
- all types of overlay data
- JPEG compressed images

Fig. 192: Dicom Display Image menu

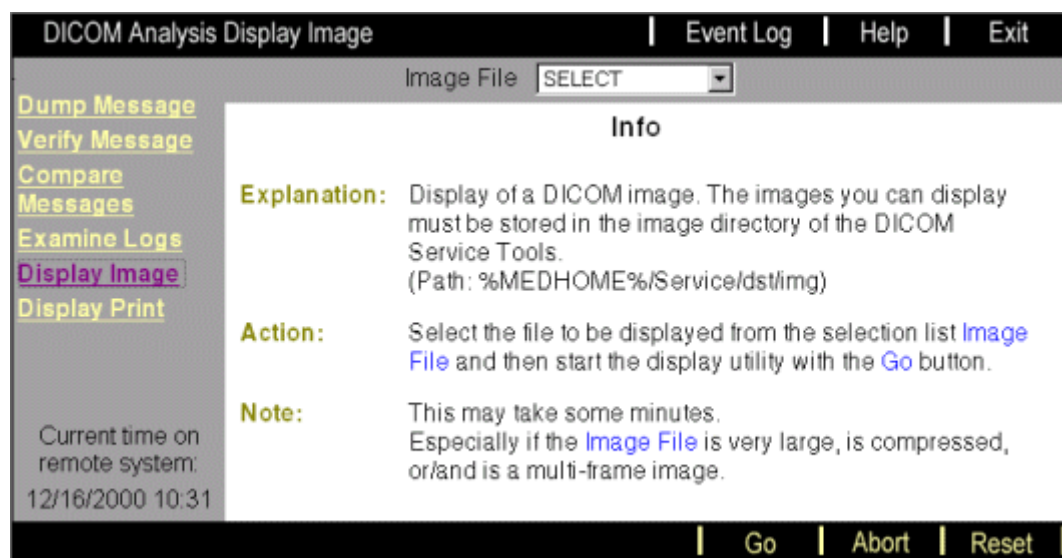
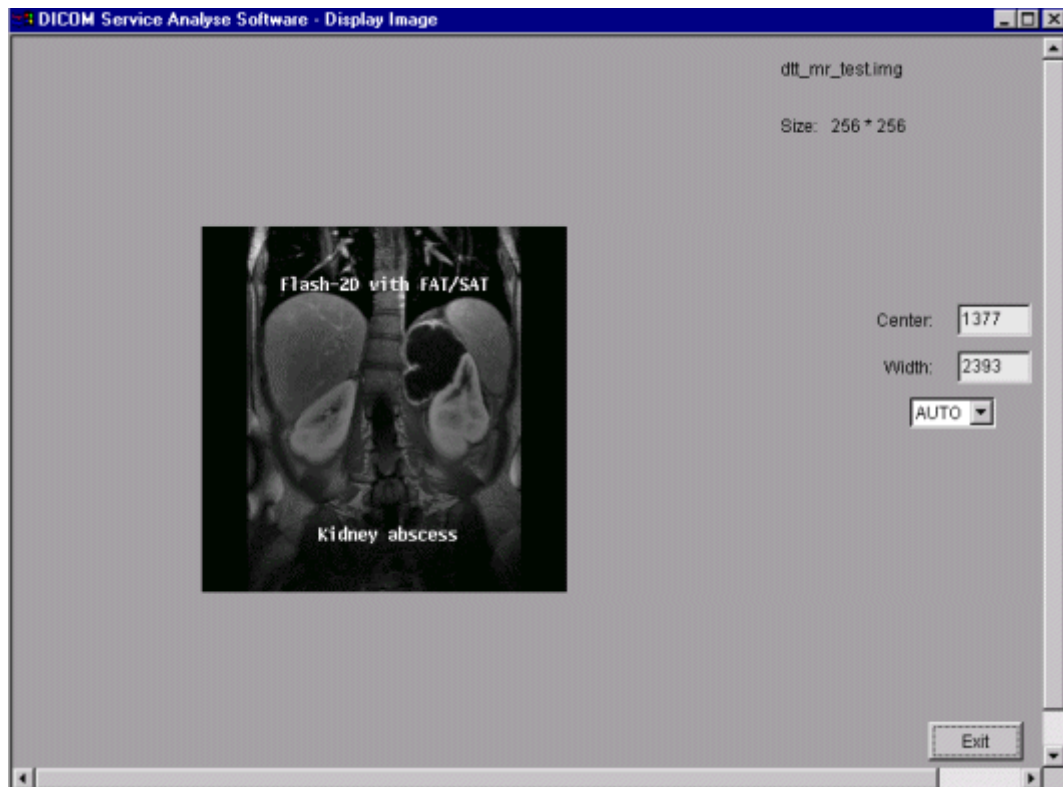


Fig. 193: The Dicom Display Image viewer



Display Print

With this option, the bitmaps generated by the camera simulator will be displayed on screen. Please note that the bitmap is exactly like the scanning array of the laser camera, i.e. the images are mirrored since the image is printed on the reverse side of the film. The pictures are in BMP (Windows bitmap) format. They are also located in the directory %MEDCOM%\service\dst\img, and may be copied to a floppy disk for further evaluation.

Fig. 194: Dicom Display Print menu



8.2.3 merge log file

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This example shows the merge_stu.log with trace level T2, T3, T7, and T9.

1. Each DICOM communication starts with the association negotiation, which may be displayed on trace level T3. The Service Class User (SCU) sends an association request message to the provider and presents all supported objects with a list of proposed transfer syntaxes (in this case only one). Each object will be identified by a Presentation Context ID. Additionally, the Application Entity Titles (AETs) and the maximum Protocol Data Unit size (PDU) will be transferred.

(312) 07-21 14:58:22.84 MC3 T3: **REQUEST Association Parameters:**

(312) 07-21 14:58:22.84 MC3 T3: | Calling_presentation_address: **cmv10001**
(146.254.120.75)

(312) 07-21 14:58:22.85 MC3 T3: | Called_presentation_address:

(312) 07-21 14:58:22.85 MC3 T3: | **Called_application_title: CSYNGO1**

(312) 07-21 14:58:22.85 MC3 T3: | **Calling_application_title: 049SA1DCDADRSP**

(312) 07-21 14:58:22.85 MC3 T3: | Application_context_name:

1.2.840.10008.3.1.1.1

(312) 07-21 14:58:22.85 MC3 T3: | **PDU_maximum_length: 28672**

(312) 07-21 14:58:22.85 MC3 T3: | Implem_class_uid: 1.3.12.2.1107.5.8

(312) 07-21 14:58:22.85 MC3 T3: | Version_name: SNKIT_1.10

(312) 07-21 14:58:22.85 MC3 T3: | Max_operations_invoked: Not Sent

(312) 07-21 14:58:22.85 MC3 T3: | Max_operations_performed: Not Sent

(312) 07-21 14:58:22.85 MC3 T3: | **PRESENTATION CONTEXTS:**

(312) 07-21 14:58:22.86 MC3 T3: | **(1)Pres_context_id: 1**

(312) 07-21 14:58:22.86 MC3 T3: | **(1)Abstract_syntax:**

1.2.840.10008.5.1.4.1.1.4 (STANDARD_MR)

(312) 07-21 14:58:22.86 MC3 T3: | **(1) Proposed transfer syntaxes:**

(312) 07-21 14:58:22.86 MC3 T3: | **1.2.840.10008.1.2 (Implicit Little Endian)**

(312) 07-21 14:58:22.86 MC3 T3: | **(1)SCU Role: Not sent (requestor supports SCU)**

(312) 07-21 14:58:22.86 MC3 T3: | **(1)SCP Role: Not sent (requestor supports SCP)**

(312) 07-21 14:58:22.86 MC3 T3: | **(1)No extended application info**

(312) 07-21 14:58:22.86 MC3 T3: | **(2)Pres_context_id: 3**

(312) 07-21 14:58:22.86 MC3 T3: | **(2)Abstract_syntax:**

1.2.840.10008.5.1.4.1.1.1 (STANDARD_CR)

(312) 07-21 14:58:22.87 MC3 T3: | **(2)Proposed transfer syntaxes:**

(312) 07-21 14:58:22.87 MC3 T3: | **1.2.840.10008.1.2 (Implicit Little Endian)**

(312) 07-21 14:58:22.87 MC3 T3: | (2)SCU Role: Not sent (requestor supports SCU)

(312) 07-21 14:58:22.87 MC3 T3: | (2) SCP Role: Not sent (requestor supports SCP)

(312) 07-21 14:58:22.87 MC3 T3: | (2)No extended application info

(312) 07-21 14:58:22.87 MC3 T3: | **(3)Pres_context_id: 5**

.....

.....

.....

(312) 07-21 14:58:23.01 MC3 T3: | **(21)Pres_context_id: 41**

(312) 07-21 14:58:23.02 MC3 T3: | (21)Abstract_syntax:

1.2.840.10008.5.1.4.1.1.1.2.1 (STANDARD_MG_PROCESS)

(312) 07-21 14:58:23.02 MC3 T3: | (21)Proposed transfer syntaxes:

(312) 07-21 14:58:23.02 MC3 T3: | **1.2.840.10008.1.2 (Implicit Little Endian)**

(312) 07-21 14:58:23.02 MC3 T3: | (21)SCU Role: Not sent (requestor supports SCU)

(312) 07-21 14:58:23.02 MC3 T3: | (21) SCP Role: Not sent (requestor supports SCP)

(312) 07-21 14:58:23.02 MC3 T3: | (21)No extended application info

2. The response of the Service Class Provider shows the objects are supported (accepted) and the proposed transfer syntaxes are used.

(312) 07-21 14:58:23.02 MC3 T3: **RESPONSE Association Parameters:**

(312) 07-21 14:58:23.02 MC3 T3: | Calling_presentation_address: cmv10001
(146.254.120.75)

(312) 07-21 14:58:23.02 MC3 T3: | Called_presentation_address:

(312) 07-21 14:58:23.02 MC3 T3: | **Called_application_title: CSYNGO1**

(312) 07-21 14:58:23.02 MC3 T3: | **Calling_application_title: 049SA1DCDADRSP**

(312) 07-21 14:58:23.03 MC3 T3: | Application_context_name:

1.2.840.10008.3.1.1.1

(312) 07-21 14:58:23.03 MC3 T3: | **PDU_maximum_length: 28672**

(312) 07-21 14:58:23.03 MC3 T3: | Implem_class_uid: 1.3.12.2.1107.5.9.990801

(312) 07-21 14:58:23.03 MC3 T3: | Version_name: SIEMENS_SWFVA47A

(312) 07-21 14:58:23.03 MC3 T3: | Max_operations_invoked: Not Sent

(312) 07-21 14:58:23.03 MC3 T3: | Max_operations_performed: Not Sent

(312) 07-21 14:58:23.03 MC3 T3: | **PRESENTATION CONTEXTS:**

(312) 07-21 14:58:23.03 MC3 T3: | **(1)Pres_context_id: 1**

(312) 07-21 14:58:23.03 MC3 T3: | (1)Abstract_syntax:

1.2.840.10008.5.1.4.1.1.4 (STANDARD_MR)

(312) 07-21 14:58:23.03 MC3 T3: | **(1)Result_transfer_syntax:**

1.2.840.10008.1.2 (Implicit Little Endian)

(312) 07-21 14:58:23.04 MC3 T3: | (1) SCU Role: Not sent (remote accepts requestor SCU role)

(312) 07-21 14:58:23.04 MC3 T3: | (1) SCP Role: Not sent (remote accepts requestor SCP role)

(312) 07-21 14:58:23.04 MC3 T3: | (1) No extended application info

(312) 07-21 14:58:23.04 MC3 T3: | **(1) Negotiation Result: Accepted**

(312) 07-21 14:58:23.04 MC3 T3: | **(2) Pres_context_id: 3**

(312) 07-21 14:58:23.04 MC3 T3: | (2) Abstract_syntax:

1.2.840.10008.5.1.4.1.1.1 (STANDARD_CR)

(312) 07-21 14:58:23.04 MC3 T3: | (2) Result_transfer_syntax:

1.2.840.10008.1.2 (Implicit Little Endian)

(312) 07-21 14:58:23.04 MC3 T3: | (2) SCU Role: Not sent (remote accepts requestor SCU role)

(312) 07-21 14:58:23.05 MC3 T3: | (2) SCP Role: Not sent (remote accepts requestor SCP role)

(312) 07-21 14:58:23.05 MC3 T3: | (2) No extended application info

(312) 07-21 14:58:23.05 MC3 T3: | **(2) Negotiation Result: Accepted**

(312) 07-21 14:58:23.05 MC3 T3: | **(3) Pres_context_id: 5**

.....

.....

.....

(312) 07-21 14:58:23.20 MC3 T3: | **(21) Pres_context_id: 41**

(312) 07-21 14:58:23.20 MC3 T3: | (21) Abstract_syntax:

1.2.840.10008.5.1.4.1.1.1.2.1 (STANDARD_MG_PROCESS)

(312) 07-21 14:58:23.20 MC3 T3: | **(21) Result_transfer_syntax: (Implicit Big Endian)**

(312) 07-21 14:58:23.20 MC3 T3: | (21) SCU Role: Not sent (remote accepts requestor SCU role)

(312) 07-21 14:58:23.20 MC3 T3: | (21) SCP Role: Not sent (remote accepts requestor SCP role)

(312) 07-21 14:58:23.20 MC3 T3: | (21) No extended application info

(312) 07-21 14:58:23.20 MC3 T3: | **(21) Negotiation Result: REJECTED: Abstract syntax**

3. After the association negotiation, the "Command Set" of the first object (e.g. image) will be sent and may be displayed on trace level T7. All attributes within the Command Set belong to group 0000 and indicate which type of object will be sent (context ID and Affected SOP Class UID) and the unique object identification (Affected SOP Instance UID)

(409) 07-21 14:58:23.34 MC3 T7: **Message received on context 5**

(409) 07-21 14:58:23.34 MC3 T7: Group 0x0000 Elements:

(409) 07-21 14:58:23.34 MC3 T7: (0000,0000) Group 0000 Length

(409) 07-21 14:58:23.34 MC3 T7: (1): 126

(409) 07-21 14:58:23.34 MC3 T7: **(0000,0002) Affected SOP Class UID**

(409) 07-21 14:58:23.34 MC3 T7: **(1): 1.2.840.10008.5.1.4.1.1.2**

(409) 07-21 14:58:23.34 MC3 T7: (0000,0100) Command Field

(409) 07-21 14:58:23.34 MC3 T7: (1): 1

(409) 07-21 14:58:23.35 MC3 T7: (0000,0110) Message ID

(409) 07-21 14:58:23.35 MC3 T7: (1): 0

(409) 07-21 14:58:23.35 MC3 T7: (0000,0700) Priority

(409) 07-21 14:58:23.35 MC3 T7: (1): 0

(409) 07-21 14:58:23.35 MC3 T7: (0000,0800) Data Set Type

(409) 07-21 14:58:23.35 MC3 T7: (1): 0

(409) 07-21 14:58:23.35 MC3 T7: (0000,1000) Affected SOP Instance UID

(409) 07-21 14:58:23.35 MC3 T7: (1):

1.3.12.2.1107.5.1.3.24013.4.0.2508718952428

4. On trace level T2, the "Command Set" with the additional information from the message ID is displayed again. This ID shows the transfer syntax used, where

100 is equal to Implicit Value Representation, Little Endian

101 is equal to Explicit Value Representation, Little Endian

102 is equal to Explicit Value Representation, Big Endian

104 is equal to RLE

105 is equal to JPEG Baseline (lossy)

106 is equal to JPEG Extended_2_4 (lossy)

122 is equal to JPEG_Lossless_Hier_14

After the Command Set, the "Data Set" will be shown including all other attributes of the sent object. In the case of image transfer, these attributes are identical to the image header information. In the case of a Query or Worklist request, you can see the attributes sent within these requests.

The value length will be shown within () and the value itself within | |.

Command Set:

(409) 07-21 14:58:26.37 MC3 T2: **Message (STANDARD_CT,C_STORE_RQ) received on context 5**

(409) **Message ID: 100**

(409) **Service Supported: STANDARD_CT (0032)**

(409) **Command Supported: C_STORE_RQ**

(409) **Message Attributes:**

(409) 0000,0000 Group 0000 Length VR: UL VM: 1

(409) (00000004) |126|

(409) **0000,0002 Affected SOP Class UID VR: UI VM: 1**

(409) (00000025) |**1.2.840.10008.5.1.4.1.1.2**|

(409) 0000,0100 Command Field VR: US VM: 1

(409) (0000002) |1|
 (409) 0000,0110 Message ID VR: US VM: 1
 (409) (0000002) |0|
 (409) 0000,0700 Priority VR: US VM: 1
 (409) (0000002) |0|
 (409) 0000,0800 Data Set Type VR: US VM: 1
 (409) (0000002) |0|
 (409) **0000,1000 Affected SOP Instance UID** VR: UI VM: 1
 (409) (0000043) |1.3.12.2.1107.5.1.3.24013.4.0.2508718952428|

Data Set:

(409) **0008,0008 Image Type** VR: CS VM: 1-n
 (409) (0000035) |ORIGINAL|
 (409) |PRIMARY|
 (409) |LOCALIZER|
 (409) |CT_SOM5 TOP|
 (409) **0008,0016 SOP Class UID** VR: UI VM: 1
 (409) (0000025) |1.2.840.10008.5.1.4.1.1.2|
 (409) **0008,0018 SOP Instance UID** VR: UI VM: 1
 (409) (0000043) |1.3.12.2.1107.5.1.3.24013.4.0.2508718952428|
 (409) 0008,0020 Study Date VR: DA VM: 1
 (409) (0000008) |20000711|
 (409) 0008,0021 Series Date VR: DA VM: 1
 (409) (0000008) |20000711|
 (409) 0008,0022 Acquisition Date VR: DA VM: 1
 (409) (0000008) |20000711|
 (409) 0008,0023 Image Date VR: DA VM: 1
 (409) (0000008) |20000711|
 (409) 0008,0030 Study Time VR: TM VM: 1
 (409) (0000013) |143100.140000|
 (409) 0008,0031 Series Time VR: TM VM: 1
 (409) (0000013) |143145.375000|
 (409) 0008,0032 Acquisition Time VR: TM VM: 1
 (409) (0000013) |143203.890001|
 (409) 0008,0033 Image Time VR: TM VM: 1
 (409) (0000013) |143203.890001|
 (409) **0008,0050 Accession Number** VR: SH VM: 1
 (409) (0000010) |FONo200012 |
 (409) **0008,0060 Modality** VR: CS VM: 1

(409) (00000002) **[CT]**
 (409) 0008,0070 Manufacturer VR: LO VM: 1
 (409) (00000007) **[SIEMENS]**
 (409) 0008,0080 Institution Name VR: LO VM: 1
 (409) (00000011) **[Siemens CTE]**
 (409) 0008,0081 Institution Address VR: ST VM: 1
 (409) (00000032) **[Street\x0A/27E5E2/\x0ADistrict\x0ACountry]**
 (409) 0008,0090 Referring Physician's Name VR: PN VM: 1
 (409) (00000061)
[RefFamNam^RefGiveNam^RefMidNam^RefTitle^RefSuf^Ref]
 (409) **[Pre^RefDegr]**
 (409) 0008,1010 Station Name VR: SH VM: 1
 (409) (00000007) **[F24013C]**
 (409) **0008,1030 Study Description** VR: LO VM: 1
 (409) (00000020) **[Abdomen^AbdThinSlice]**
 (409) 0008,103E Series Description VR: LO VM: 1
 (409) (00000008) **[Topogram]**
 (409) 0008,1040 Institutional Department Name VR: LO VM: 1
 (409) (00000000) **<null>**
 (409) 0008,1050 Performing Physician's Name VR: PN VM: 1-n
 (409) (00000047) **[SchedPerfPhNam1^SchedPerfPhNam2^SchedPerfPhNam3]**
 (409) 0008,1060 Name of Physician(s) Reading Study VR: PN VM: 1-n
 (409) (00000000) **<null>**
 (409) 0008,1070 Operators' Name VR: PN VM: 1-n
 (409) (00000012) **[Dr. Gruening]**
 (409) 0008,1080 Admitting Diagnosis Description VR: LO VM: 1-n
 (409) (00000015) **[LWS Fehlerbild1]**
 (409) **0008,1090 Manufacturer's Model Name** VR: LO VM: 1
 (409) (00000011) **[Volume Zoom]**
 (409) 0008,2111 Derivation Description VR: ST VM: 1
 (409) (00000000) **<null>**
 (409) **0010,0010 Patient's Name** VR: PN VM: 1
 (409) (00000040) **[CCTJul2000^VZVA20A^SpineSSD^Dr.^CCC_(\xFF\xFF)]**
 (409) **0010,0020 Patient ID** VR: LO VM: 1
 (409) (00000009) **[PatID_VZ3]**
 (409) 0010,0030 Patient's Birth Date VR: DA VM: 1
 (409) (00000008) **[19550824]**
 (409) 0010,0032 Patient's Birth Time VR: TM VM: 1

(409) (00000000) <null>
 (409) 0010,0040 Patient's Sex VR: CS VM: 1
 (409) (00000001) |F|
 (409) 0010,1000 Other Patient IDs VR: LO VM: 1-n
 (409) (00000000) <null>
 (409) 0010,1001 Other Patient Names VR: PN VM: 1-n
 (409) (00000000) <null>

 (409) 0018,0010 Contrast/Bolus Agent VR: LO VM: 1
 (409) (00000000) <null>
 (409) **0018,0015 Body Part Examined** VR: CS VM: 1
 (409) (00000007) |ABDOMEN|

 (409) **0018,1000 Device Serial Number** VR: LO VM: 1
 (409) (00000000) <null>
 (409) **0018,1020 Software Version(s)** VR: LO VM: 1-n
 (409) (00000007) |VA20A.1|

 (409) **0018,5100 Patient Position** VR: CS VM: 1
 (409) (00000003) |HFS|
 (409) **0020,000D Study Instance UID** VR: UI VM: 1
 (409) (00000043) |1.3.12.2.1107.5.1.3.24013.4.0.2883410072770|
 (409) **0020,000E Series Instance UID** VR: UI VM: 1
 (409) (00000043) |1.3.12.2.1107.5.1.3.24013.4.0.2882929363340|
 (409) **0020,0010 Study ID** VR: SH VM: 1
 (409) (00000001) |1|
 (409) **0020,0011 Series Number** VR: IS VM: 1
 (409) (00000001) |1|
 (409) 0020,0012 Acquisition Number VR: IS VM: 1
 (409) (00000001) |1|
 (409) **0020,0013 Image Number** VR: IS VM: 1
 (409) (00000001) |1|
 (409) 0020,0032 Image Position (Patient) VR: DS VM: 3
 (409) (00000048) | 0|
 (409) | -398|
 (409) | -1|
 (409) 0020,0037 Image Orientation (Patient) VR: DS VM: 6

(409) (0000096) | 0|
 (409) | 1|
 (409) | 0|
 (409) | 0|
 (409) | 0|
 (409) | -1|
 (409) **0020,0052 Frame of Reference UID** VR: UI VM: 1
 (409) (0000043) | **1.3.12.2.1107.5.1.3.24013.4.0.2507320363132**|
 (409) 0020,0060 Laterality VR: CS VM: 1
 (409) (00000000) <null>
 (409) 0020,1002 Images in Acquisition VR: IS VM: 1
 (409) (00000001) | 0|
 (409) 0020,1040 Position Reference Indicator VR: LO VM: 1
 (409) (00000000) <null>
 (409) 0020,1041 Slice Location VR: DS VM: 1
 (409) (0000016) | -1|
 (409) **0020,4000 Image Comments** VR: LT VM: 1
 (409) (00000000) <null>
 (409) 0028,0002 Samples per Pixel VR: US VM: 1
 (409) (00000002) | 1|
 (409) **0028,0004 Photometric Interpretation** VR: CS VM: 1
 (409) (0000011) | **MONOCHROME2**|
 (409) **0028,0010 Rows** VR: US VM: 1
 (409) (00000002) | **512**|
 (409) **0028,0011 Columns** VR: US VM: 1
 (409) (00000002) | **512**|
 (409) **0028,0030 Pixel Spacing** VR: DS VM: 2
 (409) (0000032) | 1|
 (409) | 1|
 (409) **0028,0100 Bits Allocated** VR: US VM: 1
 (409) (00000002) | **16**|
 (409) **0028,0101 Bits Stored** VR: US VM: 1
 (409) (00000002) | **12**|
 (409) **0028,0102 High Bit** VR: US VM: 1
 (409) (00000002) | **11**|
 (409) 0028,0103 Pixel Representation VR: US VM: 1
 (409) (00000002) | 0|

(409) **0028,1050 Window Center** VR: DS VM: 1-n

(409) (0000016) | **50|**

(409) **0028,1051 Window Width** VR: DS VM: 1-n

(409) (0000016) | **350|**

(409) **0028,1052 Rescale Intercept** VR: DS VM: 1

(409) (0000016) | **-1024|**

(409) **0028,1053 Rescale Slope** VR: DS VM: 1

(409) (0000016) | **1.0002441|**

.....

(409) **7FE0,0010 Pixel Data** VR: OW VM: 1

(409) <Callback routine>

(409) ===== End of Message ID: 100 =====

5. After each object (image), a response will be sent from the provider, in which you can see the status of the transfer (0 = successful). A table of the status codes is available in the Intranet at <http://dicom.med.siemens.de> at DICOM - Glossary&Tables.

This "Command Set" message may be displayed on T7.

(409) 07-21 14:58:29.85 MC3 T7: **Message (STANDARD_CT,C_STORE_RSP) sent on context 5**

(409) 07-21 14:58:29.85 MC3 T7: Group 0x0000 Elements:

(409) 07-21 14:58:29.86 MC3 T7: (0000,0000) Group 0000 Length

(409) 07-21 14:58:29.86 MC3 T7: (1): 126

(409) 07-21 14:58:29.86 MC3 T7: **(0000,0002) Affected SOP Class UID**

(409) 07-21 14:58:29.86 MC3 T7: **(1): 1.2.840.10008.5.1.4.1.1.2**

(409) 07-21 14:58:29.86 MC3 T7: (0000,0100) Command Field

(409) 07-21 14:58:29.86 MC3 T7: (1): 32769

(409) 07-21 14:58:29.86 MC3 T7: (0000,0120) Message ID Being Responded To

(409) 07-21 14:58:29.87 MC3 T7: (1): 0

(409) 07-21 14:58:29.87 MC3 T7: (0000,0800) Data Set Type

(409) 07-21 14:58:29.87 MC3 T7: (1): 257

(409) 07-21 14:58:29.87 MC3 T7: **(0000,0900) Status**

(409) 07-21 14:58:29.87 MC3 T7: (1): 0

(409) 07-21 14:58:29.87 MC3 T7: **(0000,1000) Affected SOP Instance UID**

(409) 07-21 14:58:29.87 MC3 T7: (1):

1.3.12.2.1107.5.1.3.24013.4.0.2508718952428

6. When you turn on trace level T9, you will also see the whole association on the Protocol Data Unit level (PDU). This will add a lot more information into the log file which is helpful in some special cases. For example, when the transfer is interrupted you can see how many packets have been transferred successfully.

The Association Release request and response can also only be seen on trace level T9.

(409) 07-21 14:58:29.96 MC3 **T9: PDU Type: 05 (Assoc_Rel_RQ) Received**

(409) 07-21 14:58:29.96 MC3 T9: Reserved Byte: 00

(409) 07-21 14:58:29.96 MC3 T9: PDU length: 0000 0004 (4 bytes)

(409) 07-21 14:58:29.96 MC3 T9: Reserved Bytes (7-10): 0000 0000

(409) 07-21 14:58:29.96 MC3 **T9: PDU Type: 06 (Assoc_Rel_RP) Sent**

(409) 07-21 14:58:29.96 MC3 T9: Reserved Byte: 00

(409) 07-21 14:58:29.96 MC3 T9: PDU length: 0000 0004 (4 bytes)